RF Prior: Context-Preserving Representation Adaptation for Instance Segmentation Transfer

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Abstract

Instance segmentation transfer requires learning target boundaries while retaining context encoded by a pretrained model. We introduce RF PRIOR*, a receptive-field (RF)-constrained framework that couples scale-selective preservation with a trainable contextual readout. A scale-selective parameter constraint preserves source-trained P5 weights as upstream P3/P4 features adapt along its gradient path. Pooled P5 context guides P4 through area-restricted attention (AA), followed by local P3 refinement. On three sidewalk targets, this design improves mask mean average precision (mAP) over scratch and full fine-tuning at 25.2 frames per second (FPS). Across three targets and seeds, attention reduces P4 displacement and late gradients as P3 adapts and P5 features evolve. Shared-head losses, including distribution focal loss (DFL), connect this scale-specific adaptation to mask and localization fitting. These results connect contextual reuse to preserved parameters and their target-adaptive readout.*

The project page is available at https://e2emd.github.io/rf-prior/.

1. Introduction

Transferring pretrained features to instance segmentation changes the spatial evidence required for prediction. Target masks demand both contextual recognition and precise boundaries, while changes in label inventory and annotation geometry reshape the supervision across scales [27, 53]. Because a feature pyramid distributes these cues across receptive fields [32], transfer poses a coupled question: which source parameters should be preserved, and how should their features interact with target-adaptive spatial detail?

Full fine-tuning, starting-point ℓ_2 regularization (L2-SP), and selective tuning regulate departure from source parameters [21, 23]. Yet parameter control leaves the interaction between evolving features and downstream fusion unspecified. Frozen-backbone detection establishes readout capacity as a key factor [43]. Dense transfer therefore requires coupling preservation to a decoder that integrates retained context with target boundary and foreground representations.



Figure 1. **Adaptation behavior and measured dynamics.** (a) Early-adaptation illustration; mid training is an intermediate target-adaptation stage. (b) Recorded P4 displacement, averaged over three seeds: AA reduces final movement by 19.2%/40.6%/22.4% versus w/o AA. The reduction holds across all nine matched target–seed pairs at the 480-update endpoint.

RF PRIOR couples preserved P5 parameters to trainable P3/P4 in a You Only Look Once (YOLO)v12-X pyramid [41]. As P5 inputs evolve, gradients traverse the preserved transformation and couple it to target learning. Spatial pyramid pooling-fast (SPPF) exposes contextual scales for AA to integrate with P4, after which P3 refines local mask geometry.

We evaluate this design on SideGuide's POLYSEG/SURFACEMASK targets [34] and FIXED OBJECTS, which expose changes in boundary supervision and label inventory. Three-target diagnostics connect these results to parameter movement and feature evolution. Our contribution is a preservation–readout formulation that separates fixed parameters from evolving representations, together with stage-resolved evidence linking this distinction to dense transfer.

2. Related Work

Parameter preservation and feature adaptation. L2-SP anchors source weights; Learning without Forgetting (LwF) and elastic weight consolidation (EWC) constrain source behavior or important parameters [19, 23, 25]. Spot-

Tune selects fine-tuning paths per input, and surgical fine-tuning relates layer choice to shift type [7, 21]. Residual adapters, visual prompts, and Rein adapt features around preserved backbones [13, 37, 51]; weight-decomposed low-rank adaptation (DoRA) separates magnitude and direction [28]. These works establish that fixed weights support changing representations. We specialize this principle to a spatial hierarchy: preserved wide-context transformations receive adaptive inputs, and the P5-to-P4 readout shapes local-stage gradients. Stage-resolved diagnostics distinguish this coupling from weight anchoring.

Dense transfer across tasks and scenes. Feature reuse depends on task distance, depth, and spatial support [32, 53]. Where frozen-backbone detection emphasizes readout capacity [43], our lower pyramid also adapts to target geometry. SideGuide's box and polygon annotations expose supervision and inventory shifts within related sidewalk scenes [34], complementing street-scene benchmarks [4, 33, 54] and synthetic-to-real or geographic adaptation [9, 11, 44]. Penn-Fudan pedestrians and the Wine Grape Instance Segmentation Dataset (WGISD) extend these diagnostics across target geometries [39, 48].

Contextual readout for dense prediction. Fully convolutional and pyramid models combine spatial scales [27, 30, 56]; Mask region-based convolutional neural networks (Mask R-CNN), You Only Look At CoefficienTs (YOLACT), and Segmenting Objects by Locations v2 (SOLOv2) realize different mask readouts [1, 8, 50]. Non-local, criss-cross, and transformer operators expand spatial interaction [2, 10, 49, 58]. SegFormer, Mask2Former, MaskDINO, and OneFormer develop hierarchical or masked reasoning [3, 12, 22, 29, 52]; YOLO variants emphasize real-time prediction [38, 45–47]. OneFormer with a large shifted-window Transformer backbone (OneFormer-Swin-L) supplies our accuracy reference. Our P5-to-P4 interface couples efficient cross-scale processing [57] to preserved context, followed by lightweight P3 mask refinement.

Automatic box-to-polygon priors. Box-supervised segmentation derives mask supervision through proposals, structured teachers, or consistency losses [5, 17, 20, 42]. Our offline conversion uses Segment Anything Model (SAM) proposals [18], intersection-over-union (IoU) gating, clipping, and contour simplification [6, 36]. Every FIXED OBJECTS variant uses these polygons, holding supervision fixed across parameter and readout designs.

3. Methodology

Preliminaries. Given $x \in \mathbb{R}^{3\times H\times W}$, the backbone $B_{\boldsymbol{\theta}_{\text{bb}}}$ produces a feature pyramid $\{F_3, F_4, F_5\}$ at strides $\{8, 16, 32\}$. Blocks $\boldsymbol{\theta}_{\text{bb}} = [\boldsymbol{\theta}_3; \boldsymbol{\theta}_4; \boldsymbol{\theta}_5]$ end at P3/P4/P5 with nominal receptive-field scales $R_3 < R_4 < R_5$. The head $H_{\boldsymbol{\theta}_{\text{seg}}}$ fuses these features into instance masks. Detector parameters $\boldsymbol{\theta}_0$ are learned on bounding-box source $\mathcal{D}_{\text{bb}}$ (*BBox-DS*); $\mathcal{D}_{\text{seg}} \in \{\text{POLYSEG}, \text{SURFACEMASK}, \text{FIXED OBJECTS}\}$ supplies target polygons.

Definition 1 (Scale-selective preservation)**.** Let $\boldsymbol{\Theta} = [\boldsymbol{\theta}_3; \boldsymbol{\theta}_4; \boldsymbol{\theta}_5]$ and $\mathcal{C} = \{\boldsymbol{\Theta} : \boldsymbol{\theta}_5 = \boldsymbol{\theta}_{0,5}\}$. On $\mathcal{C}$, P3/P4 and the readout remain trainable; P5 inputs and normalization state evolve under preserved learned parameters.

Cross-context objective. Set $\boldsymbol{m} = [\mathbf{0}; \mathbf{0}; \boldsymbol{\theta}_{0,5}]$ and block precision $\boldsymbol{\Lambda}_\tau = \text{blkdiag}(\xi_3 I, \xi_4 I, \tau I)$. For $\xi_3, \xi_4, \tau > 0$, the Gaussian family

$$p_\tau(\boldsymbol{\Theta}) \propto \exp\left[-\tfrac{1}{2}(\boldsymbol{\Theta} - \boldsymbol{m})^\top \boldsymbol{\Lambda}_\tau (\boldsymbol{\Theta} - \boldsymbol{m})\right], \tag{1}$$

converges as $\tau \to \infty$ to P3/P4 Gaussian shrinkage and $\delta(\boldsymbol{\theta}_5 - \boldsymbol{\theta}_{0,5})$. For targets $\mathcal{D}_{\text{seg}} = \{(x_i, y_i)\}_{i=1}^N$ and task loss in Eq. (9), the cross-context penalized risk is

$$\begin{aligned}\mathcal{L}_{\text{CC}}(\boldsymbol{\theta}) = &\sum_i \ell_{\text{task}}\big(H_{\boldsymbol{\theta}_{\text{seg}}} \circ B_{\boldsymbol{\theta}_{\text{bb}}}(x_i), y_i\big) \\ &+ \frac{\xi_3}{2}\|\boldsymbol{\theta}_3\|_2^2 + \frac{\xi_4}{2}\|\boldsymbol{\theta}_4\|_2^2, \quad \boldsymbol{\theta}_5 = \boldsymbol{\theta}_{0,5}.\end{aligned} \tag{2}$$

L2-SP instead anchors the source vector $\boldsymbol{\Theta}_0$ through

$$\Omega_{\text{SP}} = \tfrac{\lambda}{2}\|\boldsymbol{\Theta} - \boldsymbol{\Theta}_0\|_2^2, \quad \nabla\Omega_{\text{SP}} = \lambda(\boldsymbol{\Theta} - \boldsymbol{\Theta}_0), \tag{3}$$

which restores source proximity while allowing every block to adapt [23]. For stochastic gradient descent from $\boldsymbol{\Theta}^t \in \mathcal{C}$, set $g^t = \nabla_{\boldsymbol{\Theta}}\ell^t + \boldsymbol{\Lambda}_0\boldsymbol{\Theta}^t$ and $\boldsymbol{\Lambda}_0 = \text{blkdiag}(\xi_3 I, \xi_4 I, 0)$. Definition 1 gives

$$\begin{aligned}\boldsymbol{\Theta}^{t+1} &= \Pi_{\mathcal{C}}(\boldsymbol{\Theta}^t - \eta_t g^t) \\ &= \boldsymbol{\Theta}^t - \eta_t P_{\mathcal{T}} g^t.\end{aligned} \tag{4}$$

Here $\Pi_{\mathcal{C}}(u) = \boldsymbol{m} + P_{\mathcal{T}} u$, with $P_{\mathcal{T}} = \text{blkdiag}(I, I, 0)$, removes the P5 update direction. The same constraint applies to the AdamW displacements in our 480-update stud-

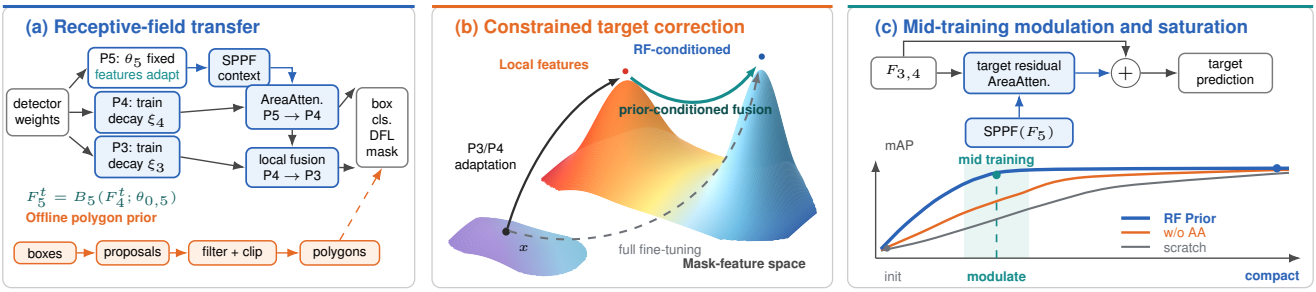


Figure 2. **Overview of RF Prior.** (a) P5 preserves source-trained parameters while its feature maps evolve with the trainable input representation. P3/P4 receive weight decay, and gradients pass through P5. (b) Conceptual mask-feature landscape: SPPF/AA provides prior-conditioned fusion; surface height is illustrative. (c) Schematic mid-training modulation and saturation: adapted context can facilitate subsequent refinement. Figure 4 offers a qualitative view of retained source-conditioned structure after Fixed Objects adaptation.

ies [31].

Global-to-local gradient transmission. Write $f_\ell = \mathrm{vec}(F_\ell)$, $J_{ab} = \partial f_a / \partial f_b$, and $J_{\ell\theta} = \partial f_\ell / \partial \boldsymbol{\theta}_\ell$. Let g_ℓ be the loss gradient at each separate readout input and $h_4 = g_4 + J_{54}^\top g_5$.

Proposition 1 (Frozen-stage gradient transmission)**.** *At a differentiable point, retaining P5 in the computation graph under $\mathcal{C}$ yields*

$$\begin{aligned} \nabla_{\boldsymbol{\theta}_4} \ell &= J_{4\theta}^\top \big(g_4 + J_{54}^\top g_5\big), \\ \nabla_{\boldsymbol{\theta}_3} \ell &= J_{3\theta}^\top \big(g_3 + J_{43}^\top h_4\big). \end{aligned} \tag{5}$$

The contextual gradient term $J_{4\theta}^\top J_{54}^\top g_5$ is retained even though the P5 parameter update is zero.

The readout thus shapes both stages through distinct Jacobians, derived in the supplement.

Vectorize spatial features as $X_\ell \in \mathbb{R}^{n_\ell \times c_\ell}$. Resampling $S_{5\to\ell} \in \mathbb{R}^{n_\ell \times n_5}$ and channel maps $\Phi = \{W_{5\ell}, W_\ell\}_{\ell\in\{3,4\}}$ give a common d-channel basis [27]:

$$\begin{aligned} R_{\ell,i} &= S_{5\to\ell} X_{5,i} W_{5\ell} - X_{\ell,i} W_\ell, \\ E_{\mathrm{align}}(\Phi) &= \sum_{i,\ell} \|R_{\ell,i}\|_F^2 = \sum_{i,\ell} \mathrm{tr}(R_{\ell,i}^\top R_{\ell,i}). \end{aligned} \tag{6}$$

For $\ell \in \{3,4\}$, the trace separates feature energies from cross-scale association at specified projections; task supervision learns the fusion.

Multi-scale attentive decoder. Let $P_0 = X_5 W_c$ and apply stride-one 5×5 max pooling $\mathcal{M}_5$ on the underlying spatial grid, returning vectorized features:

$$\begin{aligned} P_j &= \mathcal{M}_5^j(X_5 W_c), \quad k_j = 4j+1, \\ \widetilde{X}_5 &= [P_0 \| P_1 \| P_2 \| P_3] W_s \in \mathbb{R}^{n_5 \times d_s}. \end{aligned} \tag{7}$$

For $j \in \{1,2,3\}$, k_j gives the 5/9/13 interior pooling widths; $W_s \in \mathbb{R}^{4d_c \times d_s}$ fuses these contexts [14]. We upsample $\widetilde{X}_5$ and form the stride-16 matrix $U_4 = [X_4 W_4 \,\|\, S_{5\to4}\widetilde{X}_5]$. Following the area-restricted interaction used by YOLOv12 [41], for region index set $\mathcal{I}_r$ attention computes

$$\begin{aligned} [Q_r \,\|\, K_r \,\|\, V_r] &= U_4[\mathcal{I}_r][W_Q \,\|\, W_K \,\|\, W_V], \\ A_r = \mathrm{softmax}(Q_r K_r^\top / \sqrt{d}), &\quad Z_r = A_r V_r. \end{aligned} \tag{8}$$

Reassembled outputs $\{Z_r W_O\}_{r=1}^a$ enter a P4 residual projection. Context shapes both affinities A_r and values V_r, hence Proposition 1's gradient; a two-convolution cross-stage-partial (C2f)-style block then connects P4 to P3. For n stride-16 tokens, a equal regions reduce attention cost from $O(n^2 d)$ to $O(n^2 d/a)$. The w/o-AA ablation retains P5 preservation and SPPF with local fusion; plain "P5 Frozen" retains the original neck.

Unified optimization. The YOLO-style multitask loss combines complete intersection over union (CIoU), binary cross-entropy (BCE), distribution focal loss (DFL), and mask supervision [1, 24, 59]:

$$\mathcal{L} = \lambda_{\mathrm{box}}\mathcal{L}_{\mathrm{CIoU}} + \lambda_{\mathrm{cls}}\mathcal{L}_{\mathrm{BCE}} + \lambda_{\mathrm{dfl}}\mathcal{L}_{\mathrm{DFL}} + \lambda_{\mathrm{mask}}\mathcal{L}_{\mathrm{mask}}. \tag{9}$$

With area-normalized BCE over matched instance pixels, mask supervision contributes alongside the other heads at shared features, coupling localization, class separation, and discrete offset distributions to the contextual readout.

Polygon-prior mining. For proposal M_k and source box B, define $q_k = \mathrm{IoU}(M_k, B)$ and the accepted set $\mathcal{G}_\tau = \{k : q_k \geq \tau\}$. The path used to create Fixed Objects is

$$M^\star = \begin{cases} \bigvee_{k\in\mathcal{G}_\tau} M_k, & \mathcal{G}_\tau \neq \varnothing, \\ M_{\arg\max_k q_k}, & \text{otherwise}, \end{cases} \tag{10}$$

SAM supplies box-prompted candidates [18]. We simplify the union's largest external contour, clamp vertices to B,

and use the rectangle for an empty contour. Mining precedes optimization: the retained labels are held fixed and shared by every variant. Table 1(b) reports original-label visual consistency across the retained validation archive and proposal-mask consistency on 100 images.

4. Experiments

Datasets and setup. *BBox-DS* supplies the detector prior; POLYSEG changes box supervision to polygons, SURFACEMASK introduces surface-centric labels, and the smaller FIXED OBJECTS set emphasizes stationary infrastructure (Table 1). Main YOLOv12-X runs use `optimizer=auto` and mixed precision on NVIDIA RTX A6000 (POLYSEG, SURFACEMASK) or A100 (40 GB; FIXED OBJECTS) GPUs. Table 2 FPS is measured on the A100. Evaluation reports precision, recall, and mean average precision (mAP) for boxes and masks; $\text{mAP}_{50:95}$ averages IoU from 0.50 to 0.95 in 0.05 increments. Matched variants share target data, augmentations, evaluation, and schedule; only source initialization, parameter constraint or regularization, and decoder fusion change. The standard SideGuide comparison uses 30 epochs and the extended row uses 50 epochs. For FIXED OBJECTS, the reported budgets are 16.98k and 50.94k mini-batch steps, with batch size 8 and 1,698 steps per epoch. Subscripts mark second-best recorded checkpoints; budget labels follow the archived convention. Figure 5(d) separately pairs the FIXED OBJECTS logs at identical recorded epochs.

Definition 2 (Stage-wise adaptation profile)**.** For nonzero initial weights, define displacement $\Delta_\ell^\theta(t) = \|\boldsymbol{\theta}_\ell^t - \boldsymbol{\theta}_\ell^0\|_2 / \|\boldsymbol{\theta}_\ell^0\|_2$. Pre-clipping gradient root mean square (RMS) is $G_\ell(t) = \|\nabla_{\boldsymbol{\theta}_\ell} \ell^t\|_2 / \sqrt{d_\ell}$ for $d_\ell > 0$ recorded elements. Feature distance $D_\ell^{\text{feat}}(t)$ averages $1 - \cos(\bar{F}_{\ell,i}^t, \bar{F}_{\ell,i}^0)$ over fixed images, using pooled, vectorized features.

Testing preservation and readout. Scratch, full fine-tuning, plain P5 freezing, and L2-SP test initialization and Definition 1's parameter constraint; full/w/o-AA decoders test the readout under fixed P5. Definition 2 links these controls to stage-wise dynamics, while YOLO/OneFormer references and shared MobileSAM candidates [55] support accuracy–throughput and label-consistency comparisons.

Transfer accuracy and adaptation budget. Table 2 evaluates the coupled preservation–readout design against L2-SP [23] and established segmentation references [12]. On FIXED OBJECTS, the full decoder exceeds the strongest non-proposed result in each column by 1.87 box mAP_{50}, 2.69 mask mAP_{50}, 1.84 box $\text{mAP}_{50:95}$, and 2.19 mask $\text{mAP}_{50:95}$. RF PRIOR is strongest on POLYSEG, improving mask mAP_{50} by 3.56 points over L2-SP and 4.17 points over full fine-tuning. On SURFACEMASK, YOLOv9-E leads coarse mAP_{50}, whereas RF PRIOR leads strict $\text{mAP}_{50:95}$, indicating a strict-localization advantage for already plausible surfaces. The OneFormer-Swin-L reference is competitive but slower: RF PRIOR is 3.4× faster (25.2 vs. 7.4 FPS). Against No Pretraining, it raises compute from 162 to 393 GFLOPs (2.43×) while FPS decreases by 2.3%; SRI 2.34 reflects realized arithmetic throughput, and GFLOPs expose the higher arithmetic cost. The gains are largest for POLYSEG boundaries and FIXED OBJECTS masks at limited budgets.

On SideGuide, the 30-epoch comparison in Table 3 gives RF PRIOR the best recall and box/mask mAP among matched YOLOv12-X variants. Plain P5 freezing improves strict localization over full fine-tuning but remains below scratch; the complete decoder closes that gap, showing the importance of reading out the preserved context. The 50-epoch row improves every box and mask endpoint on the same held-out validation split.

Contextual readout across training budgets. SPPF exposes several context radii; AA weights their fusion with P4, whereas RF PRIOR (w/o AA) mixes them locally [14, 41]. At the archived early checkpoint in Table 4, the full decoder improves over scratch by +3.62/+4.44 box/mask mAP_{50} and +2.67/+3.26 box/mask $\text{mAP}_{50:95}$. The lead over RF PRIOR (w/o AA) is 2.18/1.31 box/mask mAP_{50} and 1.64/0.73 strict mAP. At the extended checkpoint, w/o AA leads mask mAP_{50}/$\text{mAP}_{50:95}$ by 0.46/0.18 points, while full retains a 0.29-point strict-box edge. To examine this budget dependence, paired-epoch diagnostics track optimization under the same supervision.

Qualitative protocols. Figure 3 compares direct-target P3–P5 feature maps; coral guides mark matched annotation coordinates. In contrast, Figure 4 follows *BBox-DS* pretraining and FIXED OBJECTS adaptation; scratch uses FIXED OBJECTS alone. Its cross-target POLYSEG/SURFACEMASK rows examine retained source-conditioned structure after the label-inventory change, while the self-target FIXED OBJECTS row examines spatial support for labeled objects within the target-adapted feature maps.

Shared-head optimization and accuracy. Figure 5(d) pairs validation mask-loss reduction and mask $\text{mAP}_{50:95}$ difference at 33 epochs: Pearson $r = 0.694$ overall and $r = 0.635$ after warm-up. The full decoder lowers mask loss in 25 epochs and improves strict mask mAP in nine, distinguishing pixelwise fitting from instance ranking.

Distributional localization provides a complementary view of the same adapted features. DFL learns discrete box-offset distributions [24]; mask and box heads

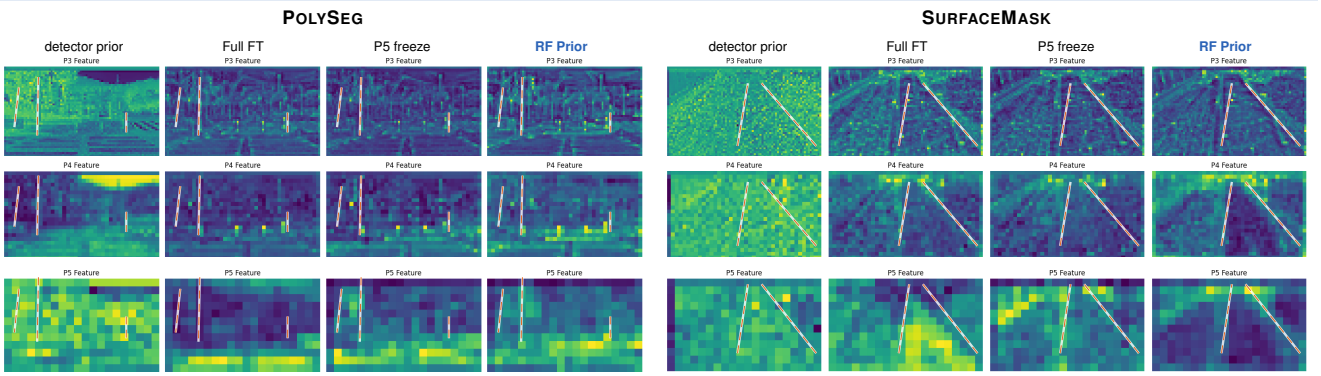


Figure 3. **Direct-target P3–P5 feature maps.** The detector prior precedes adaptation; full fine-tuning (Full FT), P5 freeze, and RF PRIOR use the pictured target. Coral guides reproject Figure 4 input geometry: tree support, signal mast, and curb bollard for POLYSEG; green-strip boundaries for SURFACEMASK, at corresponding spatial positions across the three pyramid levels for matched spatial comparison.

Table 1. **Datasets and FIXED OBJECTS label audits.** (a) BG/LS: background/label-space shift from *BBox-DS*; ▲ mild, ✓ clear. (b) MobileSAM proxy-mask overlap on 100 screened images (IoU: intersection over union; boundary F1: boundary precision–recall harmonic mean; DP: Douglas–Peucker). Final row: AI visual consistency / unresolved / flagged rates (%) for 1,599 retained images.

(a) Dataset composition

Datasets (#images)	Dataset Type	classes	*train*	*validation*	*total*
SIDEGUIDE [34]					
BBox-DS	Prior source	29	20,097	2,233	22,330
POLYSEG	BG ▲, LS ▲	29	66,082	7,342	73,424
SURFACEMASK	BG ✓, LS ✓	6	41,759	4,640	46,399
FIXED OBJECTS (Proposed)	BG ▲, LS ✓	15	13,577	1,543	15,120

(b) FIXED OBJECTS label audits

Selection / post-processing	Mask IoU (%)	Boundary F1 (%)
SAM confidence	85.18±15.99	72.76±25.20
IoU-selected mask	88.41±12.61	76.68±22.62
IoU + coverage	87.63±12.62	73.50±24.35
IoU-gated union	88.75±12.20	77.23±21.53
Union + clip + DP	**88.87±13.05**	**80.14±19.00**
Visual consistency (Astra)	96.1 / 3.1 / 0.8%	

Table 2. **Transfer and efficiency comparison.** Gray marks proposed configurations; bold/underline mark best/second-best values. FIXED OBJECTS reports box/mask (B/M) at the nominal 16.98k-step budget. Giga floating-point operations (GFLOPs) use one 640^2 image. The speed-retention index (SRI), $F_i G_i/(F_0 G_0)$, uses FPS F and GFLOPs G relative to the No Pretraining (BASE) configuration.

Method	Efficiency			POLYSEG mask		SURFACEMASK mask		FIXED OBJECTS B/M	
	FPS	GFLOPs	SRI	mAP_{50}	$mAP_{50:95}$	mAP_{50}	$mAP_{50:95}$	mAP_{50}	$mAP_{50:95}$
YOLOv9-E [47]	19.6	124	0.58	<u>46.56</u>	<u>27.62</u>	**75.59**	59.68	<u>73.75 / 60.89</u>	<u>53.37</u> / 41.28
YOLOv11-X [16]	**36.1**	160	1.38	46.01	27.41	74.64	59.22	73.00 / 60.14	53.03 / 40.94
OneFormer-Swin-L [12]	7.4	980	1.74	45.72	26.91	75.06	<u>59.91</u>	73.07 / 60.21	53.13 / 41.04
No Pretraining (BASE)	25.8	162	1.00	43.92	26.04	74.73	59.61	72.00 / 59.14	52.54 / 40.45
L2-SP [23]	25.8	164	1.03	44.86	26.61	74.68	59.72	73.08 / 60.41	53.18 / <u>41.52</u>
Full Fine-tuning	25.8	162	1.00	44.25	26.22	74.07	59.31	71.84 / 58.98	52.48 / 40.39
P5 Frozen, plain neck	25.9	162	1.01	42.95	25.32	74.14	59.58	71.22 / 58.36	52.17 / 40.08
RF PRIOR (w/ full decoder)	25.2	393	2.34	**48.42**	**28.25**	<u>75.15</u>	**60.15**	**75.62 / 63.58**	**55.21 / 43.71**

share the fused pyramid, so contextual refinement can affect both foreground support and localization. Where D-FINE explicitly refines distributions [35], our unchanged DFL objective measures representation-induced effects. SideGuide improves all four loss families in both phases (Figure 5(a)). Validation DFL retains $0.10815/0.39044 = 27.7\%$ of its early advantage, versus 11.7% for mask loss. Normalized train/validation geometric-mean reductions are 21.81%/16.76% for DFL and 15.94%/9.50% for mask loss; classification improves most proportionally (Figure 5(c)). On FIXED OBJECTS, mask-loss reductions remain positive in both phases, DFL validation reductions remain negative, and the box advantage changes sign (Figure 5(b)). Mean validation DFL changes by $+0.00285$ for full minus w/o AA alongside lower mask loss, indicating different loss trends across heads that share the adapted feature pyramid.

Cross-target adaptation dynamics. We transfer a YOLOv12-X sidewalk checkpoint's EMA (exponential moving average) encoder after one recorded FIXED OB-

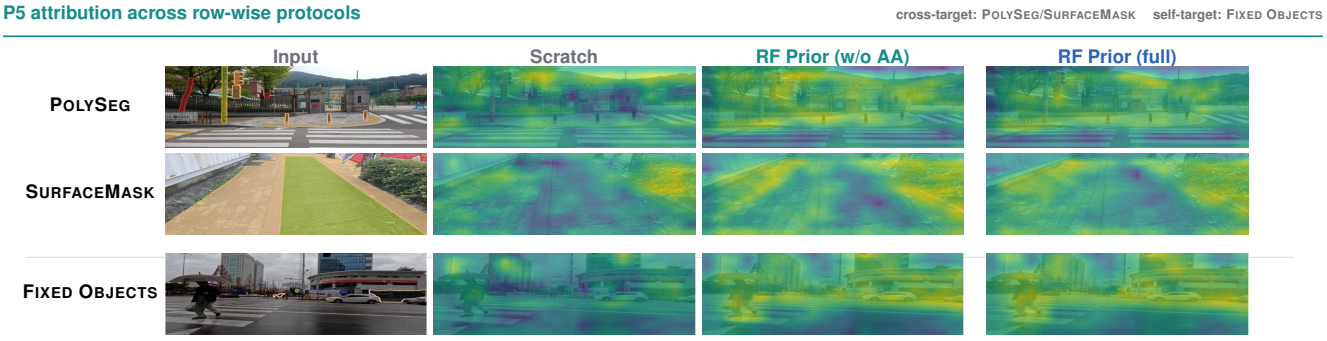


Figure 4. **Post-adaptation P5 gradient-weighted class activation mapping (Grad-CAM)** [40]. Prior-based variants follow *BBox-DS* pretraining and FIXED OBJECTS adaptation; scratch uses FIXED OBJECTS alone. POLYSEG/SURFACEMASK probe retained source-conditioned structure across targets, while FIXED OBJECTS probes self-target object focus. The retained structure motivates Figure 2(c); each row compares attribution on one spatially matched input across models trained under the corresponding adaptation protocol.

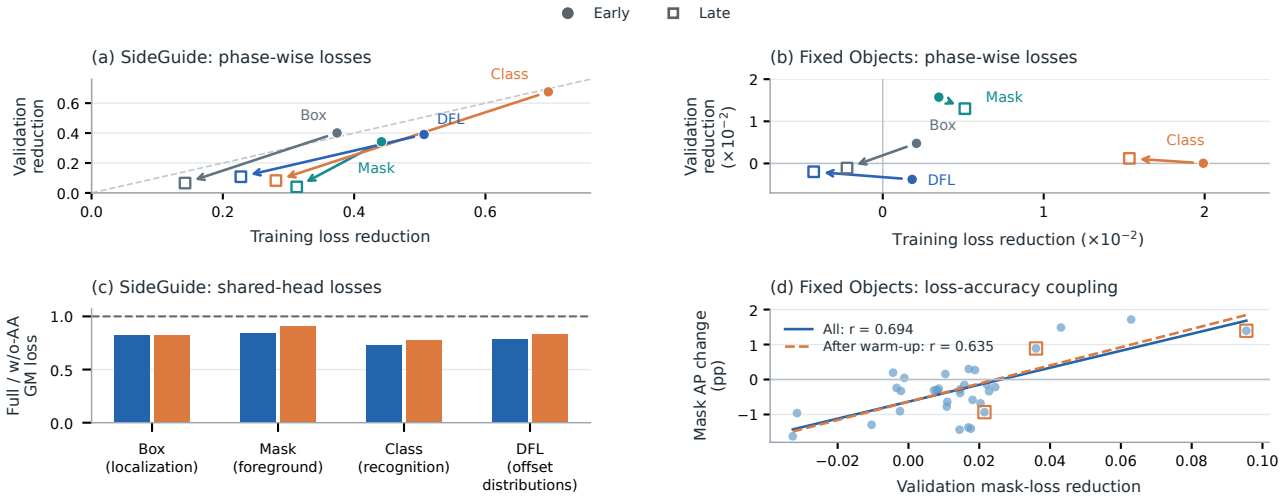


Figure 5. **Complementary optimization diagnostics.** (a,b) Mean train/validation loss reductions (w/o AA minus full), from early circles to late squares: SideGuide splits at 149.8k/299.6k steps; FIXED OBJECTS at 27.17k/56.03k mini-batches. (c) SideGuide geometric-mean (GM) loss ratios: blue training, orange validation. (d) Epoch-matched FIXED OBJECTS mask-loss reduction versus mask AP difference in percentage points (pp); squares mark warm-up epochs 1–3 and lines summarize the observed association across 33 paired epochs.

Table 3. **SideGuide validation.** P/R denote precision/recall and B/M box/mask; subscripts mark second-best checkpoints. Gray marks proposed or extended schedules; within each schedule, bold/underline mark the strongest/second-strongest recorded validation result.

Method	P(B)	R(B)	mAP_{50}(B)	$mAP_{50:95}$(B)	P(M)	R(M)	mAP_{50}(M)	$mAP_{50:95}$(M)
YOLOv9-E [47]	**71.09**$_{70.20}$	49.85$_{50.09}$	55.17$_{54.97}$	41.91$_{41.72}$	**69.69**$_{69.49}$	46.97$_{46.64}$	51.15$_{51.30}$	33.25$_{33.15}$
YOLOv11-X [16]	66.08$_{68.82}$	47.89$_{47.68}$	53.93$_{53.47}$	40.97$_{40.58}$	67.64$_{66.95}$	45.39$_{45.59}$	50.58$_{50.15}$	33.03$_{32.83}$
No Pretraining	66.61$_{66.28}$	47.15$_{46.65}$	52.30$_{52.04}$	39.51$_{39.31}$	65.41$_{64.72}$	44.70$_{44.64}$	49.08$_{48.86}$	32.08$_{31.95}$
⌞ No Pretraining, 50 epochs	67.95$_{67.62}$	49.61$_{49.95}$	54.72$_{54.57}$	41.84$_{41.71}$	69.54$_{69.27}$	45.66$_{45.83}$	51.25$_{51.12}$	33.53$_{33.47}$
Full Fine-tuning	69.38$_{68.91}$	44.97$_{44.88}$	51.53$_{51.15}$	38.78$_{38.42}$	67.72$_{66.85}$	43.11$_{43.01}$	48.30$_{47.92}$	31.48$_{31.22}$
P5 Frozen, plain neck	63.58$_{64.10}$	46.77$_{46.09}$	51.61$_{51.26}$	38.99$_{38.71}$	63.38$_{66.56}$	44.04$_{43.48}$	48.41$_{48.13}$	31.57$_{31.36}$
RF PRIOR (w/ full decoder)	70.22$_{69.48}$	**51.55**$_{51.49}$	**56.95**$_{56.63}$	**42.09**$_{41.84}$	68.08$_{67.95}$	**48.91**$_{48.08}$	**52.86**$_{52.56}$	**33.81**$_{33.71}$
⌞ **RF PRIOR (w/ full decoder), 50 epochs**	**75.84**$_{74.52}$	**56.69**$_{57.08}$	**63.61**$_{63.21}$	**48.26**$_{47.96}$	**75.11**$_{73.96}$	**52.99**$_{52.87}$	**58.98**$_{58.66}$	**37.87**$_{37.64}$

JECTS adaptation epoch, with fresh necks and heads. Its training excluded the original FIXED OBJECTS validation pool, split here into 768 adaptation/256 evaluation images; Penn-Fudan and WGISD use 120/50 and 110/27 [39, 48]. These short-budget studies use NVIDIA A100 GPUs, 480 AdamW updates [31], three seeds, and original-resolution mask AP; the supplement reports endpoints and hyperparameters. Full FT, P5 freezing, L2-SP, and DoRA use the

Table 4. **FIXED OBJECTS archived checkpoints.** P/R: precision/recall; B/M: box/mask. Subscripts: second-best checkpoints; gray: RF PRIOR variants. Full/w/o-AA use epochs 11/31 and 10/30 under the nominal budgets. Figure 5(d) compares matched epochs.

Budget	Method	P(B)	R(B)	mAP_{50}(B)	$mAP_{50:95}$(B)	P(M)	R(M)	mAP_{50}(M)	$mAP_{50:95}$(M)
16.98k	No Pretraining	$79.77_{76.98}$	$64.28_{62.06}$	$72.00_{68.26}$	$52.54_{49.32}$	$69.54_{66.48}$	$55.01_{54.47}$	$59.14_{57.76}$	$40.45_{38.38}$
16.98k	L2-SP [23]	$79.91_{77.43}$	$65.31_{63.52}$	$73.08_{70.14}$	$53.18_{50.21}$	$69.62_{66.91}$	$55.62_{54.93}$	$60.41_{58.86}$	$41.52_{39.74}$
16.98k	RF PRIOR (w/o AA)	$79.38_{78.14}$	$\mathbf{66.72}_{64.60}$	$73.44_{71.72}$	$53.57_{50.83}$	$\mathbf{73.46}_{70.32}$	$55.78_{55.60}$	$62.27_{60.51}$	$42.98_{40.48}$
16.98k	**RF PRIOR + SPPF + AA**	$\mathbf{80.26}_{75.58}$	$66.50_{67.02}$	$\mathbf{75.62}_{73.07}$	$\mathbf{55.21}_{53.16}$	$69.70_{66.90}$	$\mathbf{57.23}_{55.56}$	$\mathbf{63.58}_{60.45}$	$\mathbf{43.71}_{41.36}$
50.94k	No Pretraining	$\mathbf{87.41}_{86.49}$	$73.83_{74.14}$	$83.36_{83.57}$	$67.01_{66.70}$	$\mathbf{76.53}_{77.66}$	$63.64_{64.19}$	$70.30_{71.33}$	$51.23_{51.48}$
50.94k	L2-SP [23]	$85.96_{85.74}$	$76.92_{76.81}$	$84.72_{84.31}$	$67.89_{67.54}$	$74.91_{75.28}$	$65.91_{66.10}$	$71.42_{71.80}$	$52.21_{52.08}$
50.94k	RF PRIOR (w/o AA)	$83.72_{81.40}$	$\mathbf{81.29}_{81.60}$	$\mathbf{86.64}_{85.98}$	$69.55_{68.67}$	$73.54_{70.91}$	$\mathbf{70.84}_{70.92}$	$\mathbf{73.51}_{73.15}$	$\mathbf{53.86}_{53.77}$
50.94k	**RF PRIOR + SPPF + AA**	$83.99_{84.24}$	$80.13_{79.89}$	$86.48_{86.08}$	$\mathbf{69.84}_{69.38}$	$72.96_{72.81}$	$69.58_{69.19}$	$73.05_{72.64}$	$53.68_{53.47}$

Table 5. **Stage-policy and regularization controls on FIXED OBJECTS.** Three-seed means at update 480; subscripts give standard deviation (SD). $\Delta_3^\theta, \Delta_4^\theta$: relative P3/P4 weight displacement; D_5^{feat}: P5 cosine distance from initialization on 16 fixed images. †: zero imposed by freezing. Native-decoder P5 freeze–release trains P5 during updates 241–480 after the frozen phase.

Model / policy	Δ_3^θ	Δ_4^θ	D_5^{feat}
Native decoder			
Full FT	$0.237_{\pm 0.011}$	$0.212_{\pm 0.003}$	$0.745_{\pm 0.014}$
L2-SP [23]	$0.087_{\pm 0.003}$	$0.009_{\pm 0.001}$	$0.778_{\pm 0.025}$
P3 frozen	$0^\dagger$	$0.207_{\pm 0.015}$	$0.750_{\pm 0.010}$
P4 frozen	$0.233_{\pm 0.008}$	$0^\dagger$	$0.751_{\pm 0.011}$
P5 frozen	$0.227_{\pm 0.011}$	$0.191_{\pm 0.008}$	$0.670_{\pm 0.015}$
P5 freeze–release	$0.232_{\pm 0.015}$	$0.184_{\pm 0.013}$	$0.758_{\pm 0.040}$
RF decoder; P5 frozen			
RF Prior (w/o AA)	$0.242_{\pm 0.008}$	$0.186_{\pm 0.017}$	$0.663_{\pm 0.024}$
RF Prior (full)	$0.243_{\pm 0.003}$	$0.151_{\pm 0.013}$	$0.684_{\pm 0.001}$

native neck, while both RF variants freeze P5 and use SPPF; w/o AA substitutes a local C2f block for attention. Table 5 couples stage selection with representation continuity: P5-frozen variants yield feature distances of 0.663–0.684, versus 0.750/0.751 for native P3/P4 freezing and 0.758 after P5 release. Among the P4-trainable freezing controls, full RF Prior has the smallest P4 displacement (0.151). L2-SP reduces that displacement to 0.009, while P5 feature distance reaches 0.778, compared with 0.684 for full RF Prior. The regularizer anchors weights, while feature direction depends on upstream adaptation and normalization state, distinguishing parameter proximity from continuity of the target-image representations.

Observation 1 (Readout-dependent P4 adaptation)**.** Across the three targets and three seeds in the author-checkpoint study, the full RF decoder has lower final P4 displacement and late P4 gradient RMS than w/o AA, while mean P3 displacement differs by less than 0.4% on each target.

Final P4 displacements are 0.151/0.147/0.101 for FIXED OBJECTS/Penn-Fudan/WGISD, versus 0.186/0.248/0.130 without AA (Figure 1(b)). Late P4 gradient RMS (updates 241–480) falls by 12.6%/25.1%/16.1% over the matched schedules. In the six-method comparison of Figure 6, full RF Prior keeps P5 parameters fixed while feature cosine distance reaches 0.684/0.724/0.715, distinguishing parameter preservation from adaptive activations. P3 displacement is 0.243/0.246/0.194; its late gradient RMS changes by $+23.8/-13.2/+8.9\%$, contrasting with the consistent P4 reductions. On WGISD, the full decoder has the lowest mean mask loss (2.986) and DFL (1.923), improving on Full FT by 1.7% and 1.4% over the entire schedule (Figure 7). Native Full FT leads the box and classification means, whereas native P5 freezing leads final mask AP at 39.00; full RF Prior gains 0.84 AP over w/o AA. Across targets, full-minus-w/o-AA AP is $+0.35/-3.61/+0.84$; this separates the repeatable change in adaptation dynamics from the target-dependent accuracy ordering at the shared 480-update budget.

5. Discussion

Preserved parameters, adaptive representations. Under Definition 1, Proposition 1 links fixed P5 parameters to adaptive upstream features through the contextual readout. Figure 4's cross-target structure is consistent with retained source-conditioned organization after the FIXED OBJECTS label shift, while self-target attribution concentrates on labeled infrastructure. Figure 3 complements this view with direct-target spatial evidence across the same hierarchy.

Table 5 further separates weight anchoring from feature continuity: among adaptive-P4 methods, L2-SP restricts P4 movement most, whereas P5-frozen readouts yield smaller P5 feature distances. Definition 2 distinguishes gradient scale, weight displacement, and feature direction; Figure 6 also resolves rescaling. Native P5 freezing shares the parameter constraint but changes the readout; P3/P4 freezing and midpoint P5 release instead alter which stages adapt.

Where contextual readout changes adaptation. Observation 1 places the repeatable AA signature at P4, with lower displacement and late gradients alongside adaptive P3 features. This stage contrast localizes the contextual readout's measured effect within the adapting hierarchy.

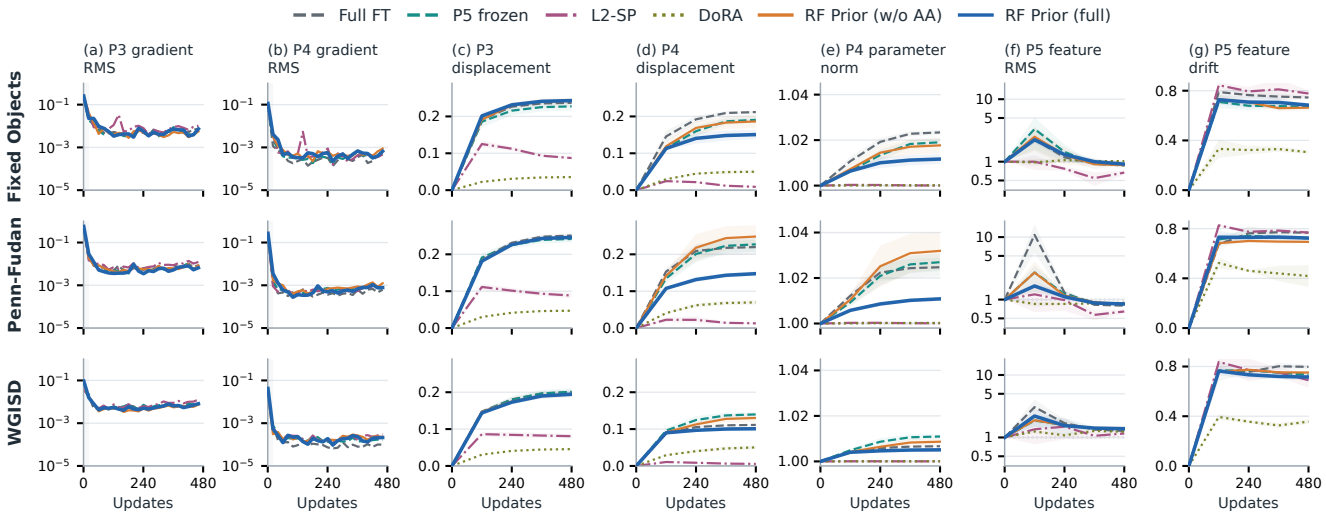


Figure 6. **Gradient, weight, and feature dynamics.** Rows are targets; columns are (a,b) pre-clipping P3/P4 gradient RMS, (c,d) P3/P4 relative displacement, (e) P4 parameter norm and (f) P5 feature RMS relative to initialization, and (g) P5 cosine distance on 16 fixed images. Lines average three seeds over optimizer updates; (c–g) ribbons show SD. Warm-up is shaded; the supplement reports DoRA adapter gradients and complete norm/feature diagnostics across all three pyramid stages over the matched optimization schedules.

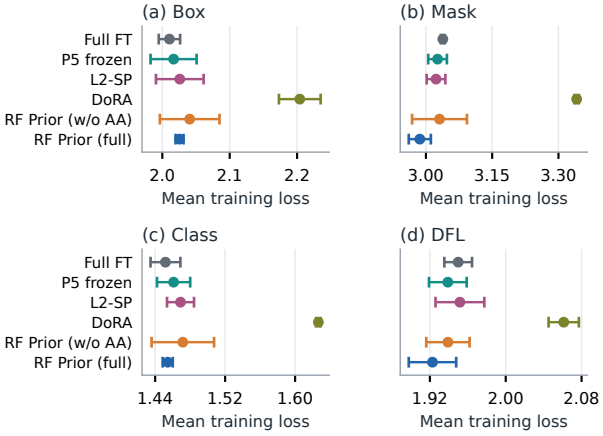


Figure 7. **WGISD shared-head fitting.** Full-schedule means for all four training losses over 480 updates. Points and bars show mean±SD across three seeds, with identical update budgets and shared target annotations for the six evaluated transfer methods.

At the task level, the PolySeg gain over full fine-tuning (4.17/2.03 mask mAP points) links the design to mask transfer, while SurfaceMask emphasizes strict localization under surface-centric supervision.

SideGuide and WGISD extend the analysis to distributional box-offset fitting [24], consistent with the adapted pyramid feeding several prediction heads. Mask and box objectives shape P5 inputs through a shared Jacobian, coupling contextual reuse to mask fitting and distributional localization under the same preserved parameter block.

Transfer across supervision and initialization. Holding supervision fixed also permits policy comparisons across target geometries [21, 53]. YOLOv12-S/YOLOv8-S encoders [15, 41] pretrained on Common Objects in Context (COCO) [26] exhibit the same P4 direction across six backbone–target means: AA reduces displacement by 0.9–7.5% and late gradient RMS by 4.5–34.7%, with P3 displacement changes below 2.8%. This repeated P4 pattern connects contextual readout to adaptation across different source initializations and model capacities.

6. Conclusion

RF Prior formulates dense transfer through scale-selective parameter preservation and a trainable contextual readout. Its gradient decomposition explains how preserved P5 parameters participate in target learning as upstream features evolve. This scale-selective design improves strict localization over full fine-tuning across three sidewalk targets at 25.2 FPS. Stage-wise diagnostics identify its characteristic behavior: attention reduces P4 movement and late gradients while P3 remains adaptive and P5 features evolve. Moreover, shared-head and cross-initialization analyses connect this adaptation pattern to different target geometries and source models. The resulting design principle couples the preserved scale to its readout, connecting constrained parameters to adaptive target representations through the gradient structure of instance segmentation under shifts in target labels and geometry.

LLM Usage Statement

OpenAI ChatGPT and Codex assisted with manuscript editing, mathematical exposition, literature search, experiment and plotting code, execution support, and project-page preparation. AI visual review was also used to assess the retained validation annotations; its protocol and image-level results are described in Table 1(b) and the supplementary material. The authors retain responsibility for the scientific claims, references, implementation, and reported results.

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